Metrogyl® Injection (5 mg/ml)

Metronidazole Injection USP
(For Intravenous administration)

COMPOSITION:
Each 100 ml of injection contains:
Metronidazole USP….. 500 mg

PRODUCT DESCRIPTION :
A clear colourless to pale yellow solution

PHARMACODYNAMICS:
Metronidazole is an effective, broad-spectrum antiprotozoal and antibacterial drug from the group of nitroimidazole. It is effective for the clinical treatment of patients with pyoinflammatory diseases.
Metronidazole is a nitro imidazole with antiprotozoal and antibacterial actions. It is highly effective against Trichomoas vaginalis. Entamoeba histolytica and Giardia lambia. It has a bactericidal action against pathogenic anaerobic bacteria, particularly Becteroides spp and Fusobacterium spp.
Bacteria inhibited by concentrations of metronidazole as low as 3.1 mcg/ml and killed by concentrations of 6.3 mcg/ml include B.fragilis, B.melaninogenious Campylobacter fetus, C.perfringes, Eubacterium, Peptococcus, Fusobacterium, Peptostreptococcus and Veilonella spp. In presence of mixed flora (both aerobes and anaerobes), metronidazole acts synergistically with antibiotics effective against common aerobic pathogens.
Metronidazole alone has no direct activity against aerobes and facultative anaerobes.

PHARMACOKINETICS:
Following an intravenous infusion (over 20 minutes) of 500 mg Metronidazole, in patients with anaerobic infections serum concentration achieved were 35.2 mcg/ml at 1 hour, 33.9 mcg/ml at 4 hour, and 25.7 mcg/ml at 8 hours.
The half life following a single intravenous dose is 6-7 hours. Metronidazole is only slightly bound to plasma proteins. It readily penetrates tissues and has a large apparent volume of distribution equivalent to distribution over 70 – 95% body weight. It attains bactericidal concentrations in most tissues and body fluids including brain, CSF, abscess cavities, saliva, bile, vaginal secretions, amniotic fluid and breast milk.

Metabolism:
Metronidazole is eliminated in man largely by metabolism resulting from side – chain oxidation, hydroxylation or conjugation of the parent compound. The major metabolic product is 1- (2-hydroxyethyl)-2 hydroxymethyl – 5 nitroimidazole which along with its glucuronide accounts for 40% to 50% of recovered urinary material. The acid and alcohol metabolites of metronidazole are 50% and 30% as active respectively as metronidazole.

Excretion:
Over a period of 24 hours urinary recovery of total nitro derivatives accounts for 35% to 65%. Renal clearance is 10.2ml per min/1.75 sq.m. After intravenous administration of a low dose 63% of the total radioactivity is excreted in the urine and 6.2% in the faeces over a 3-days period. In patients with a normal biliary tract, the concentration of metronidazole in gall bladder bile after an intravenous dose of 500mg is significantly higher than in serum.
Influence of Disease on Kinetics:
Accumulation of metronidazole has been demonstrated in serum after continual doses in a patient with impaired renal function. Therefore, dosage frequency can be reduced in patients with severe renal insufficiency.

INDICATIONS:
Metronidazole IV is an isotonic formulation, composed of a superior grade of injectable metronidazole specially developed by Unique's R&D division in India, for intravenous administration in susceptible life threatening infections.
It is particularly suited for patients who cannot retain oral medication and for patients in whom desired serum concentrations have to be attained rapidly and reliably. Metronidazole IV is thus indicated for
I. Treatment of infections caused by Bacteroides fragilis and other species of Bacteroides like Fusobacteria, Eubacteria and Anaerobic cocci such as
(a) Intra abdominal infections-appendicitis, cholecystitis, peritonitis, liver abscess and post-operative wound infection.
(b) Gynecological and obstetrical infections – puerperal sepsis, pelvic pellulitis, pelvic peritonitis.
(c) Respiratory infections- necrotising pneumonia, empyema, and lung abscess.
(d) Central nervous system injections –meningitis, brain abscess.
(e) Miscellaneous infections – septicemia, gas gangrene, osteomyelitis.

II. Prevention of post operative infections due to anaerobic bacteria.

III. Treatment of amoebic abscess of the liver and in moribund or toxic cases of fulminating intestine.

DOSAGE:
Adults and Children over 12 years of age:
500 mg (100 ml) infused over a period of 20 minutes at a rate of 5 ml/minute repeated 8 hourly.
Children below 12 years of age:
Depending upon clinical and bacteriological assessment, the physician may decide the duration of treatment. Depending on the weight of the child the volume of fluid to be infused should be determined on the basis of 7.5 mg/kg. The rate of infusion as well as the frequency remains the same as in adults: 20 minutes at the rate of 5 ml/minute repeated 8 hourly.

MODE OF ADMINISTRATION:
For IV only.

CONTRAINDICATIONS:
Metrogyl is contraindicated in patient with a prior of hypersensitivity to metronidazole or other nitromidazole derivatives.

In patient with trichomoniasis.
Metrogyl is contraindicated during the first trimester of pregnancy.

WARNING AND PRECAUTIONS:
As metronidazole is excreted in relatively high concentration in human breast milk, its use should be avoided in nursing mothers. It should be avoided, if possible, during first trimester of pregnancy, for its effects on foetal development are not definitely known. Being a 'nitro-imidazole' derivative, moderate leucopenia has been reported in some patients. It is advisable to carry out total differential leucocyte count before, during and after treatment in patients requiring long term metronidazole therapy.

Cases of severe hepatotoxicity/ acute hepatic failure, including cases with a fatal outcome with very rapid onset after treatment initiation in patients with Cockayne syndrome have been reported with products containing metronidazole for systemic use. In this population, metronidazole should therefore be used after careful benefit-risk assessment and only if no alternative treatment is available. Liver function tests must be performed just prior to the start of the therapy, throughout and after end of treatment until liver function is within normal ranges, or until the baseline values are reached. If the liver function tests become markedly elevated during treatment, the drug should be discontinued.
Patients with Cockayne syndrome should be advised to immediately report any symptoms of potential liver injury to their physician and stop taking metronidazole.

INTERACTION WITH OTHER MEDICAMENTS:
1. Clinically significant interaction between metronidazole and Warfarin has been reported.
Metronidazole has been found to increase prothrombin time. It is preferable to discontinue oral anticoagulants 24 hours prior to administration of Metrogyl I.V.
2. Simultaneous administration of metronidazole and disulfiram has been reported to cause acute psychosis and confusional state.
3. It is advisable to avoid mixing I.V. infusions of different drugs. In keeping with this rule; Metrogyl INJECTIONS should not be mixed with any other drug.

PREGNANCY AND LACTATION:
Pregnancy—Studies have not been done in humans. Metronidazole has not been shown to cause birth defects in animal studies; however, use is not recommended during the first trimester of pregnancy.
Breast-feeding—Use is not recommended in nursing mothers since metronidazole passes into the breast milk and may cause unwanted effects in the baby. However, in some infections your doctor may want you to stop breast-feeding and take this medicine for a short time. During this time the breast milk should be squeezed out or sucked out with a breast pump and thrown away. One or two days after you finish taking this medicine, you may go back to breast-feeding.

SIDE EFFECTS:
Side effects have seldom been reported in patients treated with intravenous metronidazole. Further studies are needed to determine the frequency and nature of adverse effects likely to be encountered with higher dosage and greater duration of metronidazole treatment, sometimes necessary for the treatment of severe anaerobic infections. Adverse effects which occur commonly with dosage of metronidazole used to treat trichomoniasis and amoebiasis, have included anorexia, nausea, abdominal pain, vomiting, vertigo, tiredness, and dark colouration of the urine.
Less commonly reported side effects have included ataxia, headache, transient and reversible neutropenia, metallic taste, vaginal and urethral burning, gastric irritation, diarrhoea, furred tongue. Peripheral neuropathy has also occurred in a patient treated with I.V. metronidazole for anaerobic infection. Transient epileptiform seizures have reported in a few patients undergoing intensive high dosage metronidazole treatment for radiosensitization.

SYMPTOMS AND TREATMENT OF OVERDOSE:
Three is no human experience with over dosage of Metronidazole. The acute oral toxicity of the metronidazole formulation was determined to be greater than 5g/kg (The highest dosage given) in the Albino rats.

DIRECTION FOR USE:

1. The overwrap is a moisture barrier. The inner container maintains the sterility of the product.
2. After removing overwrap, check for minute leaks by squeezing the container. Does not use if leaks are found and return for replacement.
3. Bring the injection bottle to room temperature or preferably to 37°C just before use.
4. Clean the spout of injection bottle with surgical spirit
5. Keep the bottle on a table or on a hard surface and insert the cannula of the sterile infusion set into the spout of injection bottle. (Please note that that the cannula of the infusion set should be inserted fully and not half way into the spout of the Metronidazole I.V. bottle to avoid leakage).
6. For administration hang the bottle upside down.
7. Inject intravenously slowly at a rate of about 5ml/minute.
8. To admit air, insert a sterile injection cannula on the top of the inverted bottle anywhere above the level of the liquid.

Warning: Even invisible damage to bottle caused during storage or transit, may result in contamination. Do not use if leak found on squeezing, or contents not clear, and return for replacement.

Caution: Admixture of 10% Dextrose, Penicillin G Potassium & Ringer Lactate Solution to Metrogyl Infusion is contraindicated because of chemical incompatibility.

PRESENTATION – 100ml bottle.

STERILE, NON-PYROGENIC, ISOTONIC, SINGLE DOSE CONTAINER.

STORAGE: Store below 30 °C. Do not freeze. Protect from light.

SHELF LIFE: 3 Years.

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Manufactured by:
UNIQUE PHARMACEUTICAL LABORATORIES.
(A Division of J.B. Chemicals & Pharmaceuticals Ltd.)
304 – 308, GIDC, Panoli, 394 116, India
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